



Estd. 1999

Chhatrapati Shivaji Institute of Technology

Approved by: AICTE, New Delhi | Affiliated to CSVTU, Bhilai

3.3.1 Number of research papers published per teacher in the Journals as notified on UGC CARE list during the Academic Year 2019

S.No	Title of Paper	Name of the author/s
1	Fabrication and Machinability Analysis of Squeeze Cast Al 7075/ H-Bn/Graphenehybrid Nanocomposite	Dr.Sourabh Kumar Soni
2	Establishment of Green Supply Chain management in Steel Industries Using interpretive Structural Modeling	Dr. Rajesh Kumar
3	Assessment of the Methods for the Implementation of Green Supply Chain Management in Industries	Dr. Rajesh Kumar
4	Evaluating The Performance of Plant by Overall Equipment Effectiveness & Overall Resource Effectiveness: A Case Study	Dr. K Sridhar
5	Seasonal Variation Of Carbonaceous Matter In Urban Residential Areas of Bhilai Durg Region In Chhattisgarh, India	Dr. Shailendra Kumar Kushawaha

SANTOSH
KUMAR
SHARMA

Digitally signed by SANTOSH KUMAR SHARMA
DN: C=IN, S=Chhattisgarh, Phone=
05454ef6c6bd504207baada3e7ead37aefcf4d479af531af
609675d22145a4, PostalCode=490022, STREET=Durg,
OID.2.5.4.65=c9fae72caa08a09ecf82112ae0c6c08,
SERIALNUMBER=
b0b16f4e68e16040c502d19d48f908f410b1184fac7376
6f1d8e5148c05e, O=Personal, CN=SANTOSH KUMAR
SHARMA
Reason: your signing reason here
Location: your signing location here
Date: 2024.05.16 16:17:32+05'30'
Foxit PDF Reader Version: 2023.2.0

Engineering Research Express



PAPER

Fabrication and machinability analysis of squeeze cast Al 7075/h-BN/graphene hybrid nanocomposite

RECEIVED
3 May 2019REVISED
31 May 2019ACCEPTED FOR PUBLICATION
4 June 2019PUBLISHED
11 July 2019

Parakh Agarwal, Abhimanyu Kishore, Vishnu Kumar, Sourabh Kumar Soni and Benedict Thomas

School of Mechanical Engineering, VIT, Vellore, 632014, Tamil Nadu, India

E-mail: benedictthomas4@gmail.com

Keywords: aluminium metal–matrix composites (AMMCs), graphene, h-BN, hybrid nanocomposites, ball milling, roughness, tool wear**Abstract**

This article deals with fabrication and machinability analysis of squeeze cast Al 7075/h-BN/Graphene hybrid nanocomposite (HNC), which has been fabricated by reinforcing hexagonal boron nitride (0.5 wt% h-BN) and graphene nanoparticles (1 wt% GNPs). In order to utilize the self-lubricating property of h-BN and GNPs, their uniform mixing is essential, hence before squeeze casting of HNC ball milling (BM) technique has been employed which enables uniform mixing and also eliminates the agglomeration effect of nanoparticles. Scanning electron microscopy (SEM) and optical microscopic (OM) investigation confirm the uniform mixing of nanoparticles as well as refinement in the grain size. In order to examine the hardness of the proposed HNC, mechanical properties were investigated and observed improvement of 31.25%, 10.93% and 10.27% in the UTS, microhardness (Vickers) and Rockwell hardness respectively as compared to unreinforced Al 7075 alloy fabricated by stir casting. Based on the obtained results machinability analysis is performed considering numerous machining process parameter during CNC turning to investigate the influence of cutting speed (CS), feed rate (FR) and depth of cut (DOC) on surface roughness (SR), generated forces, tool wear and chip morphology of squeeze cast HNC subjected to dry and minimum quantity lubrication (MQL) machining. Finally, the acquired results are presented with the aid of comparative graphical presentation with squeeze casted conventional aluminium alloy.

1. Introduction

In the recent era, conventional materials were substituted with lightweight aluminium metal matrix composites (AMMCs) in defence sectors, medical equipment's, sports goods, mining industries, automobiles and aircraft structures, owing to their superior mechanical characteristics such as excellent stiffness and good wear resistance [1, 2]. Numerous research articles available in the literature show that the aluminium based MMCs reinforced with silicon carbide (SiC), Titanium carbide (TiC) boron carbide (B_4C) and aluminium oxide (Al_2O_3) possess superior tensile strength, hardness and wear resistance properties. Enhanced properties of resulting composites are attained by the uniform accumulation of this micro-sized particles reinforcement in the base matrix [3–5]. Many specific applications need more superior and higher strength advanced materials, where the cost of the end product is not the constraint for their employment such as in the aerospace and automobile industries. AMMCs reinforced with nanomaterials such as carbon nanotubes (CNTs) and graphene also known as nanocomposite materials entered into the materials arena due to the technological advancement [6]. As compared to micro-sized particles reinforcements, nanometer-sized reinforcement (nanomaterial's) found more suitable for enhancing the mechanical and thermal properties of the Al-based composites and for this reason AMMCs reinforced with nanomaterials are found to be an ideal candidate for replacing the conventional materials as well as MMCs [7].

For the fabrication of such advanced nanocomposites, numerous preparation techniques such as stir casting [8, 9], squeeze casting [10], powder metallurgy [11] and friction stir processing (FSPs) [12] were frequently employed by the researcher in the huge available literature. As compared to other techniques casting (stir and squeeze) were mostly preferred owing to simple in operation, higher mechanical properties, low cost and



BRILL



Establishment of Green Supply Chain Management in Steel Industries Using Interpretive Structural Modeling

In: Perspectives on Global Development and Technology

Authors: **Rajesh Kumar** and Shiena Shekhar

[View More +](#)

Online Publication Date: 22 Oct 2019

[Download Citation](#)

[Get Permissions](#)

[Get Access](#)

Abstract

The state of Chhattisgarh in India has a very large number of steel plants causing pollution in the region. The effect of this pollution exceeds the geographical territory of a unit, and goes much beyond it, so it becomes essential to find the reasons for the pollution and the enablers for the green supply chain management, which in turn will help in providing a cleaner environment. In this study Multi-Criteria Decision Making (MCDM) tools like Interpretive Structural Modeling and MICMAC analysis have been used.

[Humanities Matter Blog](#)

[Interviews](#)

[Podcasts](#)

[Videos](#)

[Guest Posts](#)

[Podcast Series](#)

[Across the Rainbow](#)

[In Chains](#)

[Migration](#)

[Quality Education](#)

[Survival by Degrees](#)

[War and Peace](#)

**SANTOSH
KUMAR
SHARMA**

Digitally signed by SANTOSH KUMAR SHARMA
DN: C=IN, S=Chhattisgarh, Phone=+919454466666, Email=cs5454e@bcbdb504207baada3e7ead37aefcf4d479af531af60967b5d22145a4, PostalCode=490022, STREET=Durg,
OID.2.5.4.65=c9ae72caa0d8a09ecf82112ae0c6c08,
SERIALNUMBER=b001f84e68e16040f502d19d48f9908410b1184fac73766f1f8e6148c69e, O=Personal, CN=SANTOSH KUMAR SHARMA
Reason: your signing reason here
Location: your signing location here
Date: 2024.05.16 16:17:32+05'30'
Foxit PDF Reader Version: 2023.2.0

Assessment of the Methods for the Implementation of Green Supply Chain Management in Industries

Rajesh Kumar¹ and Shiena Shekhar²

¹ Mechanical Engineering Department, Bhilai Institute of Technology,
Durg, Chhattisgarh, India, PIN – 491001

² Mechanical Engineering Department, Bhilai Institute of Technology,
Durg, Chhattisgarh, India, PIN – 491001

Abstract

The generations hand over the resources and the environment of the earth to its future generation for utilization and the harnessing of the resources but over the years the under the pretext of development and the greed to earn more and in lesser time has made the mankind destroy the beautiful earth. However, efforts can be made following the Green Supply Chain Management approach so that that the production and the service activities go on as usual with least damage to the environment. In this

Keywords: Green Supply Chain Management, Green Manufacturing, Sustainable Development, Chhattisgarh

paper the authors have made efforts to understand Green Supply Chain Management and the approach to be adopted to implement it with an aim of reducing the adverse effect of the production processes used by the manufacturing industries. The effort is also made to emphasize upon the long term benefits of a Green Manufacturing process and also the important role which the industries can play in preventing further damage to the environment.

1. Introduction

The human race has an insatiable appetite for the growth and the development and it is lured by the comfort accompanying it but has been selfish enough not to look beyond the narrow window of greed and selfishness and in the run up to self satisfaction it has been the main cause for the depletion of the forest cover, soil erosion, change of weather cycle, climatic turbulence and many man-made natural disasters. The human beings have tried to give an excuse for every of its mistake which is destroying the environment and making life difficult to sustain.

The need of the hour is that the human beings should adopt the practices which are not necessary for the short term gains but the long term benefits.

The industries are the workhorses which churn out the appetizers of the human beings and they further increase its appetite. The lure of producing more at a lesser time has always captivated the human minds thus resulting in the neglect of the safety

norms and the damage to the environment. The topline and the bottomline have become the matter of prime concern and the planning and the execution of the industrial policies of the production units circumsvents around them. The atmospheric and the ground resources provided by the nature has been considered by the human beings to be perennially of a good quality and their inane production activities will not bring any harm to them but the same has been belied which the warming of the globe, the melting of the glaciers and the rising of the sea level has proven.

The fact remains that the production activities cannot be stopped or curtailed but sincere efforts should be made to identify the elements responsible for the degradation of the environment and ensure the implementation and the maintenance of sustainable technologies.

Self realization has become the matter of debate where the industry is vocal upon the implementation of the sustainable technologies but refrains from implementing it with sincerity and purpose.



EVALUATING THE PERFORMANCE OF PLANT BY OVERALL EQUIPMENT EFFECTIVENESS & OVERALL RESOURCE EFFECTIVENESS: A CASE STUDY

Rahul Pandey, **Dr. K Sridhar**

M.Tech Scholar, Department of Mechanical Engineering, C.S.I.T. Durg
Professor, Department of Mechanical Engineering, C.S.I.T. Durg

Abstract — In the highly competitive business environment, well run organizations continually strive to enhance their capabilities to create excellent value for the customers by improving the cost effectiveness of the operations. Significant improvement has taken place in the management of resources associated with manufacturing systems, to reduce the wastage of resources. The Total Productive Maintenance (TPM) concept provides a quantitative metric-Overall Equipment Effectiveness (OEE), for measuring the effectiveness of equipment or a production line. In this project, a method is developed to evaluate Overall Resource Effectiveness (ORE) by including the factors known as readiness, changeover efficiency, availability of material and availability of manpower. ORE addresses various kinds of losses associated with manufacturing system, which can be targeted for initiating improvements. Further, a case study is presented for the evaluation of ORE in a manufacturing line. Also an attempt is made implement the concept of TPM, which resulted in the improvement of machine life & reduced the failure & breakdown due to various measures taken for maintenance. The study is conducted by collecting data from Hariom Industry, Bhilai.

Keywords— Overall Resource Effectiveness, Overall Equipment Effectiveness. Total Productive Maintenance.

I. INTRODUCTION

In the present highly competitive business environment, well run organizations continually strive to enhance their capabilities to create excellent value for the customers by improving the cost effectiveness of the operations. Significant improvement has taken place in the management of resources associated with manufacturing systems, to reduce the wastage of resources. The Total Productive Maintenance (TPM) concept provides a quantitative metric-Overall Equipment Effectiveness (OEE), for measuring the effectiveness of equipment or a production line. In this study, a method is developed to evaluate Overall Resource Effectiveness (ORE) by including the factors known as readiness, changeover efficiency, availability of material and availability of manpower. ORE addresses various kinds of losses associated with manufacturing system, which can be targeted for initiating improvements. Further, a case study is presented for the evaluation of ORE in a manufacturing line.

In today's global and highly competitive environment, it is essential for the survival of any firm to be adaptive, price competitive, responsive and proactive and has the capability to deliver world class products according to diverse customer requirements. These challenges force companies to implement various Lean tools to meet the needs of the ever-changing market demand. To be a world class organization and to be stable in the global market, the firm does not have to operate worldwide or even nationwide. It may be a small local organization, which leads in its area/fields embraces and actively demonstrates to the characteristics of world-class performance. World class performance maintains, continued success through the development of an organizational environment that is distinctly different from peer and competitor firms in its philosophy and wealth-creating formula. A great number of companies find that in spite of huge improvements in productivity, there is still a bigger and better potential to utilize machine tools and reach better productivity goals. One of the main methods to meet these challenges is Total Productive Maintenance (TPM). After TPM implementation, it is necessary to monitor the overall effectiveness of the manufacturing system and benchmark it with the World class standard which will enable to continuously improve and become a World class organization.

In the last few decades industries were forced to shift their business models from closed system- orientations, to more open system-orientations. This shift was brought about by drastic competitive forces, which made the customer the focus of organizational, operational and strategic practices. Today's industries are required to operate as open operational systems. In such systems advanced operational manufacturing and process technologies are blended with modern information and communication technologies to integrate and coordinate operational resources, processes, and activities in order to generate a stream of value-added operations aimed at capturing and sustaining a competitive advantage. With the increasing complexity, scope, and organizational role of operational advanced industry technologies the maintenance of these technologies is becoming very critical to the ability of the organization to compete. In this context, maintenance management is taking on a broader organizational strategic role.

SEASONAL VARIATION OF CARBONACEOUS MATTER IN URBAN RESIDENTIAL AREAS OF BHILAI DURG REGION IN CHHATTISGARH, INDIA

Shailendra Kumar Kushawaha^{1*}, Yasmeen F. Pervez¹, Sumita Nair², Shamsh Pervez³

¹Department of Chemistry

Chhatrapati Shivaji Institute of Technology, Durg-491001 C.G

*Author for correspondence- shailendrakushawaha24@gmail.com;

²Department of Applied Chemistry

Bhilai Institute of Technology, Durg- 491001 C.G

³School of Studies in Chemistry

Pt. Ravishankar Shukla University, Raipur- 492010 C.G

Abstracts: To evaluate the carbonaceous species characteristics of PM_{2.5} in Durg District of Chhattisgarh, India, nine residential areas were selected in this study (three houses each in residential colony, near roadside, and in industrial zone), during summer and winter. The mass concentrations of PM_{2.5} has shown following trend in summer: roadside houses ($173.86 \pm 29.60 \mu\text{g}/\text{m}^3$) > industrial zone houses ($139.05 \pm 30.09 \mu\text{g}/\text{m}^3$) > residential colony houses ($133.06 \pm 31.02 \mu\text{g}/\text{m}^3$). The mass concentrations of PM_{2.5} trend obtained in winter was: roadside houses ($388.24 \pm 67.65 \mu\text{g}/\text{m}^3$) > industrial zone houses ($295.96 \pm 71.52 \mu\text{g}/\text{m}^3$) > residential colony houses ($269.24 \pm 38.77 \mu\text{g}/\text{m}^3$). It is observed higher than the annual standard stipulated by CPCB called National Ambient Air Quality Standard (NAAQS) for fine ($40 \mu\text{g}/\text{m}^3$) and by the United State Environmental Protection Agency (USEPA) (PM_{2.5} = $15 \mu\text{g}/\text{m}^3$). The mass concentration of OC has shown trend in summer: roadside houses ($61.87 \pm 11.49 \mu\text{g}/\text{m}^3$) > industrial zone houses ($46.33 \pm 10.09 \mu\text{g}/\text{m}^3$) > residential colony houses ($45.64 \pm 12.09 \mu\text{g}/\text{m}^3$). The mass concentration of OC trend obtained in winter was: roadside houses ($136.44 \pm 23.29 \mu\text{g}/\text{m}^3$) > industrial zone houses ($102.99 \pm 24.77 \mu\text{g}/\text{m}^3$) > residential colony houses ($99.84 \pm 16.16 \mu\text{g}/\text{m}^3$). The mass concentration of EC trend obtained in summer was: roadside houses ($19.53 \pm 4.44 \mu\text{g}/\text{m}^3$) > residential colony houses ($16.29 \pm 3.95 \mu\text{g}/\text{m}^3$) > industrial zone houses ($14.04 \pm 4.16 \mu\text{g}/\text{m}^3$). The mass concentration of EC trend obtained in winter was: roadside houses ($36.36 \pm 7.75 \mu\text{g}/\text{m}^3$) > industrial zone houses ($29.44 \pm 8.74 \mu\text{g}/\text{m}^3$) > residential colony houses ($28.55 \pm 4.56 \mu\text{g}/\text{m}^3$). High loads of carbonaceous particulates found in the studies suggest that carbonaceous particulate in PM_{2.5} are major respirable pollutants, threatening the health of residents in the study area in Durg district of Chhattisgarh.

Keywords: Organic carbon, Elemental carbon, PM_{2.5}, Residential areas, Fine particulate.

Introduction:

Air pollution is a global environmental burden, and has been identified as a significant Human health risk (Pant et al., 2016). Particulate matter of diameter 2.5 μm (PM_{2.5}) in aerodynamic diameter are generally considered "fine" particulates, have been implicated in adverse human health effects. Numerous epidemiological studies have shown an association between fluctuations in ambient particulate matter (PM) levels and changes in daily mortality and morbidity (Dockery and Pope, 1994; Seaton et al., 1995; Wallace, 2000). However, ambient PM levels at central monitoring sites cannot be effectively associated with community human exposure in epidemiological studies because people spend up to 90% of their time indoors (e.g. work, school, or home) (Spengler and Sexton, 1983; Wallace, 1996). Most people spend 65– 70% of their time inside their residences (Spengler and Sexton, 1983). The elderly, the very young, and those of ill health – all of whom are more susceptible to the effects of poor air